



Atsforge: An AI-Powered Career Management Platform for ats Resume Building, JD Matching, Cover Letter Generation, and Professional Outreach

Ajay Pratap Singh Yadav

Assistant Professor

Computer Science and Engineering (Artificial Intelligence)

IIMT COLLEGE OF ENGINEERING, Greater Noida, India

er.ajay17@gmail.com

Anshu Prasad

B.Tech Student

IIMT COLLEGE OF ENGINEERING, Greater Noida,

India

anshukumarprasad565@gmail.com

Deepak Yadav

B.Tech Student

IIMT COLLEGE OF ENGINEERING, Greater Noida,

India

deepak045227@gmail.com

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Abstract— Job searching has become harder than it should be. Most job seekers do not realize that their resumes are being filtered out automatically by Applicant Tracking Systems (ATS) before any human ever reads them. In this paper, we present ATsForge — a web-based platform we built to help candidates improve resumes and related job-application assets using AI-assisted workflows. The platform integrates a resume builder with live ATS scoring, a JD Match analyzer, a cover letter generator, a professional outreach suite, and a curated job portals hub. We use Groq-hosted LLaMA inference for resume parsing, job-description understanding, and content generation, enabling resume parsing in under 2 seconds. For resume evaluation, ATsForge applies a five-factor hybrid ATS scoring model based on section completeness, action and evidence, keyword spread, parse safety, and language quality. For resume-to-role comparison, the platform also applies a separate four-factor JD Match model based on weighted keyword coverage, semantic equivalence, title alignment, and critical requirement coverage. We tested the platform on 30 real job descriptions and found that the average ATS score improved by 36.3 points, while the time needed to prepare a complete application dropped from 67 minutes to under 5 minutes. <https://atsforge.co.in/>

Index Terms— Applicant Tracking System, Resume Optimization, Large Language Models, LLaMA 3.3, Groq LPU, NLP, Keyword Extraction, Cover Letter, Career Management, Serverless LaTeX, React.js, Supabase.